IT Workshop II (Cloud Computing): Services and Platforms

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Amazon Simple Notification Service

- Amazon Simple Notification Service (SNS) is a push messaging service from Amazon.
- SNS has two types of clients - publishers and subscribers.
- Publishers communicate asynchronously with subscribers by producing and sending messages to topics.
- Subscribers are the consumers who subscribe to topics to receive notifications.

Amazon Elastic Transcoder

- Amazon Elastic Transcoder is a cloud-based video transcoding service.
- Elastic Transcoder can be used to convert video files from their source format into various other formats that can be played on devices such as desktops, mobiles, tablets, etc.
- Transcoding pipelines are used to perform multiple transcodes in parallel.
- Elastic Transcoder works with the Amazon S3 storage where the input and output video files are stored.

Content Delivery Services

- Cloud-based content delivery service include Content Delivery Networks (CDNs).
- A CDN is a distributed system of servers located across multiple geographical locations to serve content to end-users with high availability and high performance.
- CDNs are useful for serving static content such as text, images, scripts, etc., and streaming media.
- CDNs have a number of edge locations deployed in multiple locations.

Amazon CloudFront

- Amazon CloudFront is a content delivery service.
- CloudFront can be used to deliver dynamic, static and streaming content using a global network of edge locations.
- The content in CloudFront is organized into distributions.
- Each distribution specifies the original location of the content to be delivered which can be an Amazon S3 bucket, an Amazon EC2 instance, an Elastic Load Balancer etc.
- CloudFront helps in improving the performance of websites.

Amazon AutoScaling

- Amazon AutoScaling allows automatically scaling of Amazon EC2 capacity up or down according to user defined conditions.
- With AutoScaling users can increase the number of EC2 instances running their applications seamlessly during spikes in the application workloads to meet the application performance requirements and scale down capacity when the workload is low to save costs.
- **AutoScaling Service:** A connection to AutoScaling service is first established by calling boto.ec2.autoscale.connect_to_region function.
- **Launch Configuration:** After connecting to AutoScaling service, a new launch configuration is created by calling conn.create_launch_configuration.

Amazon AutoScaling

- **Launch Configuration:** Launch configuration contains instructions on how to launch new instances including the AMI-ID, instance type, security groups, etc.
- **AutoScaling Group:** After creating a launch configuration, it is then associated with a new AutoScaling group. AutoScaling group is created by calling conn.create_auto_scaling_group. The settings for AutoScaling group such as the maximum and minimum number of instances in the group, the launch configuration, availability zones, optional load balancer to use with the group, etc.
- **AutoScaling Policies:** After creating an AutoScaling group, the policies for scaling up and scaling down are defined. We can define a scale up policy with adjustment type ChangeInCapacity and scaling_adjustment $= 1$

Amazon AutoScaling

- **AutoScaling Policies:** Similarly, we can define a scale down policy with adjustment type ChangeInCapacity and scaling_adjustment = -1
- **CloudWatch Alarms:** With the scaling policies defined, the next step is to create Amazon CloudWatch alarms that trigger these policies. We can define the scale up alarm using the CPU Utilization metric.

Amazon Elastic Map Reduce

- Amazon Elastic MapReduce (EMR) is the MapReduce service from Amazon based the Hadoop framework running on Amazon EC2 and Amazon S3.
- EMR supports various job types:
  - **Custom JAR:** Custom JAR job flow runs a Java program that you have uploaded to Amazon S3.
  - **Hive program:** Hive is a data warehouse system for Hadoop. You can use Hive to process data using the SQL-like language, called Hive-QL.
  - **Pig programs:** Apache Pig is a platform for analyzing large data sets that consists of a high-level language (Pig Latin) for expressing data analysis programs, coupled with infrastructure for evaluating these programs.

Amazon Elastic Beanstalk

- Amazon provides a development service called Elastic Beanstalk that allows you to quickly deploy and manage applications in the AWS cloud.
- Elastic Beanstalk supports Java, PHP, .NET, Node.js, Python, and Ruby applications.
- With Elastic Beanstalk you just need to upload the application and specify configuration settings in a simple wizard and the service automatically handles instance provisioning, server configuration, load balancing, and monitoring.
- The launch wizard allows you to specify the environment details such as name, URL, application file, container type, instance type., etc.

Amazon Cloud Formation

- Amazon CloudFormation is a deployment management service from Amazon.
- With CloudFront you can create deployments from a collection of AWS resources such as Amazon Elastic Compute Cloud, Amazon Elastic Block Store, Amazon Simple Notification Service, Elastoc Load Balancing and Auto Scaling.
- A collection of AWS resources that you want to manage together are organized into a CloudFormation stack.

The End